

Research Literature Review

SIH 2026 — Problem Statement 26059

AI-Enabled Antarctic Sea-Ice, Iceberg Trajectory, and Navigation Decision Support System

Organization: Ministry of Earth Sciences (MoES) | Department: NCPOR | Theme: Transportation & Logistics

Note on method: This review was compiled using live web search (Aug 2026) across IEEE Xplore, ScienceDirect, Springer, Nature, MDPI, AGU/Wiley, arXiv/EGUsphere preprints, and official agency portals (NASA, NSIDC, Copernicus/ESA, ECMWF, BAS, SCAR). Every paper listed below was independently found and verified through search; where a detail (e.g., an exact page number or a secondary co-author) could not be confirmed, that field is marked **"not verified — check source directly."** No DOI, dataset, or result has been invented.

1. Problem Decomposition

The PS asks for one integrated platform with three predictive/optimization engines feeding a decision-support layer. Breaking it down:

#	Component	What it requires researching
1	Sea-ice concentration (SIC) forecasting	Which DL architectures (U-Net, ConvLSTM, Transformer) work at what lead times (daily → seasonal); Antarctic vs Arctic transferability; input variables beyond raw SIC (SST, wind, pressure)
2	Sea-ice drift / motion prediction	Satellite-derived ice motion (optical flow, feature tracking on SAR/scatterometer), wind- and current-driven forcing, joint SIC+drift models
3	Iceberg detection & tracking	SAR/optical detection (CNN, U-Net, ViT+XGBoost), operational tracking databases, size/type classification, distinguishing icebergs from sea ice and ships
4	Iceberg trajectory prediction	Physical drift models (wind+current+sea-ice-locking forces) vs data-driven trajectory forecasting; validation against tracking databases
5	Oceanographic & meteorological inputs	SST, ocean currents, SSH, wind, waves — which reanalysis/forecast products exist, at what resolution, and how ML models consume them as covariates or forecast them independently
6	Route optimization	Graph search (Dijkstra/A*), metaheuristics (ACO/PSO/GA), RL-based path planning; ice-risk-aware and fuel/multi-objective formulations; POLARIS-style risk indexing
7	Decision-support / GIS integration	How prediction outputs (SIC maps, iceberg positions, risk indices) get fused into an operator-facing system; existing Antarctic-specific systems (SatSINS, RouteView) as design precedents
8	Evaluation & validation	Metrics used per task (RMSE/MAE/IEEE for SIC; EPE for drift; F1/IoU for detection; Pareto-front quality for routing) and what baselines (persistence, SEAS5, POLARIS) a student team should benchmark against

2. Research Areas (Deliverable 2)

Six literature clusters map onto the components above and structure the table in Section 3: **A.** Sea-Ice Concentration Forecasting · **B.** Sea-Ice Drift/Motion · **C.** Iceberg Detection & Tracking (incl. trajectory) · **D.** Oceanographic/Meteorological Prediction · **E.** Maritime Route Optimization · **F.** Decision Support & GIS/Risk Systems.

3. Comprehensive Research Paper Table

Columns condensed for readability; Dataset/Method detail is expanded in prose for the Top-10 (Section 4). "Antarctic?" flags papers using Antarctic data directly vs. Arctic/generic methodology that is transferable.

A. Sea-Ice Concentration Forecasting

No	Paper (Authors, Year)	Venue / DOI	Antarctic?	Method	Dataset (source, resolution, period)	Key
A1	Andersson et al., 2021 — IceNet	Nat. Commun. 12, 5124. DOI 10.1038/s41467-021-25257-4	No (Arctic)	U-Net, probabilistic (3-class SIC)	CMIP6 sims (1850–2100) + obs 1979–2011, 25 km	Beats EC on seasonal summer f

No	Paper (Authors, Year)	Venue / DOI	Antarctic?	Method	Dataset (source, resolution, period)	Key
A2	Wang, Li, Ping et al., 2026 — ASICNet	<i>Adv. Atmos. Sci.</i> 43, 578–590. DOI 10.1007/s00376-025-5024-2	Yes	Data-driven DL + novel HybridLoss (sea-ice-edge constraint)	Satellite SIC (not verified — exact product; check source)	Lower MAE (0.022) and integrated error vs. previous versions; better dynamical baselines
A3	Wang et al., 2023 — SIPNet	<i>Geophys. Res. Lett.</i> DOI 10.1029/2023GL104347	Yes	Pure DL (SIC-only input), no explicit physics	Historical SIC fields	Outperforms ECMWF/ISPEAR and Markov models (subseasonal week) Analysis
A4	ANTSIC-UNet , 2025	<i>The Cryosphere</i> 19, 6381. tc.copernicus.org/articles/19/6381/2025/	Yes	U-Net trained on multiple climate variables	Multi-variable climate fields (not verified — full variable list; check source)	Extended seasonal skill vs. SLP trend baselines
A5	"Antarctic sea ice prediction with a ConvLSTM network," 2024	<i>Ocean Modelling</i> (ScienceDirect) DOI via sciencedirect.com/science/article/abs/pii/S1463500324000738	Yes	ConvLSTM	Daily Antarctic SIC, 2018–2022	~1-month skill; best autumn/winter
A6	Frequency-Compensated Network, 2025	arXiv:2504.16745	No (Arctic)	CNN + frequency-domain compensation	Daily Arctic SIC	Improves baselines (Grigoryev daily prediction)
A7	Kvanum et al., 2024/2025 — short-term high-res forecasting	<i>The Cryosphere</i> 19, 4149. tc.copernicus.org/articles/19/4149/2025/	No (Arctic)	DL trained on ice-charts + weather forecasts	AMSR2, ice charts	Lowest normalized Integrated Error (nIE) vs. nextSIM, persistence times test
A8	Zhang et al. — Hierarchical Information-Sharing CNN	arXiv:2311.00167	No (Arctic)	CNN jointly predicting SIC + sea-ice velocity (SIV)	ERA5 (wind/temp) + Copernicus TOPAZ4 baseline	Joint SIC-velocity outperforms single-task
A9	IceDiff, 2025 (CVPR)	computer.org/csdl/proceedings-article/cvpr/2025 ; arXiv:2410.09111	No (Arctic)	ViT (coarse forecast) + diffusion model (super-resolution)	25 km → 6.25 km SIC	First DL to deliver 6.25 km resolution forecasts
A10	MSS-STFormer, 2025	<i>Cold Regions Sci./ScienceDirect</i> DOI via S092427162500468X	No (Arctic)	Multi-scale spatiotemporal Transformer, 14 env. variables, explainable (SHAP/LIME)	OSTIA + ERA5, 60-day horizon	RMSE 0.1 correlation 0.6 at 60 days
A11	GraphTransformer joint SIT+SIC, 2026	<i>J. Mar. Sci. Eng.</i> 14(5), 495. DOI 10.3390/jmse14050495	No (Arctic)	Graph-structured spatial model + temporal Transformer	Gridded SIC+SIT fields	Simultaneous SIC+thick ice forecast
A12	"Understanding the Limitations of Deep Transformer Models for Sea Ice Forecasting," 2025	Springer chapter, DOI via link.springer.com/chapter/10.1007/978-3-031-97632-2_8	No	Critical analysis of Transformer failure modes	—	Shows transformer fails to replicate annual sea ice long-range
A13	Kim et al. (cited in A8) — CNN monthly-ahead SIC	(secondary citation — not verified, locate primary)	No (Arctic)	CNN	Satellite SIC + weather (prev. months)	MAE 2.28 correlation 0.6

B. Sea-Ice Drift and Motion Prediction

No	Paper	Venue / DOI	Antarctic?	Method	Dataset	Key Result	Limitation	Relevance
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No	Paper	Venue / DOI	Antarctic?	Method	Dataset	Key Result	Limitation	Relevance
B1	Martin & Gallego, 2025 — "Towards Reliable Sea Ice Drift Estimation..."	arXiv:2510.26653 (NeurIPS 2025 workshop)	No (Arctic)	Benchmark of 48 DL optical-flow models	RADARSAT-2 ScanSAR, validated vs GNSS buoys	Several models reach sub-km accuracy (EPE 300–400 m)	Beaufort Sea study area only; needs Antarctic SAR re-validation	□□□□□
B2	Zhang et al. — Hierarchical Info-Sharing CNN (=A8)	arXiv:2311.00167	No	CNN, joint SIC+SIV	ERA5 + TOPAZ4	Improved SIV prediction via shared features with SIC	Arctic; needs Southern Ocean current product analog	□□□□
B3	Holland & Kwok, 2012 — "Wind-driven trends in Antarctic sea-ice drift"	<i>Nature Geoscience</i> 5. DOI 10.1038/ngeo1627	Yes	Statistical analysis of 19-yr satellite ice-motion record	Satellite-tracked ice motion, 1992–2010	Establishes wind forcing as dominant driver of Antarctic ice-drift/concentration trends	Purely diagnostic, not a forecasting model	□□□□□ (foundational physics)
B4	"An analytical model of iceberg drift" (Wagner et al.)	arXiv:1610.06403	Yes	Physical/analytical drift model	BYU Antarctic Iceberg Tracking DB (QuikSCAT, 1999–2009, 352 icebergs)	Validated against observed Weddell Sea / Antarctic Peninsula trajectories	Not ML-based; useful as physics baseline for hybrid models	□□□□

C. Iceberg Detection, Tracking & Trajectory

No	Paper	Venue / DOI	Antarctic?	Method	Dataset	Key Result
C1	Braakmann-Folgmann et al., 2023 — "Mapping the extent of giant Antarctic icebergs with Deep Learning"	<i>The Cryosphere</i> 17, 4675–4690. DOI 10.5194/tc-17-4675-2023	Yes	U-Net binary segmentation	Sentinel-1 SAR, 191 images / 6 named giant icebergs; code+data on Zenodo (10.5281/zenodo.7875599)	F1 = 0.84, median error 4.1%; 0.01 s/image vs. minutes manually
C2	Iceberg detection from Sentinel-1 WV-mode SAR with YOLO, 2026	<i>Remote Sens. Environ.</i> (ScienceDirect) DOI via S0034425726001148	Yes	YOLO detector + WV-Net self-supervised foundation model	Sentinel-1 WV mode (global); iceberg dataset on Zenodo 10.5281/zenodo.17165466	Builds toward long-term SAR iceberg database
C3	Machine-Learning Detection of Icebergs in Sea Ice/Open Water, 2025	<i>Remote Sensing</i> 17(4), 702. DOI 10.3390/rs17040702	No (E. Canada)	ViT features + climate params → XGBoost classifier	RADARSAT Constellation Mission (RCM), 2022–23, C-band dual-pol	96.5% overall accuracy, 1% false positive rate
C4	Zhan et al., 2018 — "Iceberg-Ship Detection with SAR"	arXiv:1812.07367	No	CNN transfer learning	Statoil/C-CORE Kaggle SAR dataset	Log-loss 0.1463 in competition
C5	Ship-Iceberg Classification in SAR & Multispectral Images, 2020	<i>Remote Sensing</i> 12(15), 2353. mdpi.com/2072-4292/12/15/2353	No	CNN (2 architectures compared)	SAR + multispectral	Comparable accuracy across network datasets
C6	Budge & Long, 2018 — "A Comprehensive Database for Antarctic Iceberg Tracking Using Scatterometer Data"	<i>IEEE JSTARS</i> 11(2). DOI 10.1109/JSTARS.2017.2784186	Yes	Scatterometer-based tracking + interpolation, size/rotation estimation	BYU/NIC consolidated DB, 1978, 1992–present	Foundational operational Antarctic iceberg dataset
C7	"A six-year circum-Antarctic icebergs dataset (2018–2023)"	<i>Earth Syst. Sci. Data</i> 18, 147 (2026). essd.copernicus.org/articles/18/147/2026/	Yes	Semi-automated incremental random-forest classification	Sentinel-1 via Google Earth Engine, south of 55°S, Oct. snapshots 2018–23	First unified circum-Antarctic multi-scale iceberg dataset

No	Paper	Venue / DOI	Antarctic?	Method	Dataset	Key Result
C8	Modeling giant-iceberg drift w/ sea-ice influence, Weddell Sea	<i>J. Glaciology</i> (Cambridge Core)	Yes	Physical model with added "sea-ice locking force" term	Iceberg C-7 trajectory, 2+ years	Wind+current alone insufficient; sea-ice concentration/thickness term needed for realistic drift
C9	Wesche & Dierking framework (12-yr Antarctic iceberg-evolution simulation)	(secondary citation via ResearchGate figure — not verified, locate primary 2015 paper)	Yes	Size-class drift+melt simulation	BYU DB (QuikSCAT, 1999–2009) + GPS buoys	Differentiates drift by 5 size classes

D. Oceanographic and Meteorological Prediction

No	Paper	Venue / DOI	Method	Dataset
D1	Forecasting SST using ML, 2025	<i>J. Earth Syst. Sci.</i> DOI 10.1007/s12040-024-02483-0	Comparative ML/DL survey	Arabian Sea, Bay of Bengal, S. Indian Ocean SST, 1–7 day lead
D2	SEA-ViT, 2024 arXiv:2409.16313		ViT + BiGRU spatio-temporal covariance	30-yr HF radar surface currents (Gulf of Thailand/Andaman) + ENSO indices
D3	Physics-guided SSTFormer, 2026	ScienceDirect DOI via S1674987126000095	Physics-guided Transformer	Global SST
D4	DL correction of global ocean forecasts, South China Sea, 2026	<i>Front. Mar. Sci.</i> DOI 10.3389/fmars.2026.1826293	BiLSTM/ConvGRU bias-correction of GOFS output (XiHe backbone)	SST/SSS/currents/SSI state variables
D5	Hersbach et al., 2020 — ERA5	ECMWF/Copernicus C3S (climate.copernicus.eu)	Reanalysis (not ML, but the standard ML input product)	Global, 0.25°, hourly, 1940–present
D6	Copernicus Marine GLORYS12V1 (Global Ocean Physics Reanalysis)	data.marine.copernicus.eu/product/GLOBAL_MULTIYEAR_PHY_001_030	NEMO ocean model + 3D-Var/Kalman-filter assimilation	1/12°, 50 levels, 1993–present; includes SIC/SIT/ice velocity

E. Maritime Route Optimization

No	Paper	Venue / DOI	Method	Dataset/Context	Key Result	Limitation	Relevance
E1	"Optimal ice routing of a ship with icebreaker assistance," 2019	<i>Ocean Engineering</i> (ScienceDirect) DOI via S0141118718307806	3D Ant Colony Algorithm (3D-ACA), multi-objective	Time-varying ice-field grid	Jointly optimizes fuel consumption + navigation risk, with icebreaker-assist option	Simulation-based; no live satellite ice feed	□□□□□

No	Paper	Venue / DOI	Method	Dataset/Context	Key Result	Limitation	Relevance
E2	Guinness et al., 2014 — "A Method for Ice-aware Maritime Route Optimization"	IEEE PLANS 2014. DOI 10.1109/plans.2014.6851512	Waypoint/speed optimization	Ice-covered route grid	Foundational ice-aware routing formulation	Older (2014); superseded by later dynamic methods	□□□
E3	"Arctic weather routing: a review of ship performance models and ice routing algorithms," 2023	Front. Mar. Sci. DOI 10.3389/fmars.2023.1190164	Review/comparison —		Systematic comparison of ship-performance + ice-routing model pairs; identifies research directions	Review only, no new model	□□□□ (best "Phase 1 read")
E4	Flexible Velocity Obstacle (FVO) algorithm for dynamic sea-ice, 2024	Ocean Engineering DOI via S0029801824021814	FVO (velocity-obstacle robotics method adapted to ice)	Drifting-ice-zone simulation	Real-time maneuvering guidance, avoids large floe cores	Local/reactive; needs global planner on top	□□□□
E5	AHPSOGA waypoint optimization for Arctic Great Ellipse Routes, 2025	J. Mar. Sci. Eng. 13(8), 1543. DOI 10.3390/jmse13081543	Adaptive hybrid PSO-GA + POLARIS risk model	Historical/recommended NEP routes + remote-sensing ice data	16–19% faster convergence vs. plain GA/PSO; risk-comparable to real NEP route	Arctic (NEP), not Antarctic	□□□□
E6	RouteView, 2022 / RouteView 2.0, 2025	Int. J. Digital Earth DOI 10.1080/17538947.2022.2126016 ; Environ. Model. Softw. DOI via S1364815225001483	Improved A* on layered dynamic-risk graph	Big-Earth-data pipeline: SIC, SIT, temperature along NEP	Operational, interpretable route-planning + escort-demand assessment tool	Northeast Passage (Arctic); direct architectural template for an Antarctic analog	□□□□□ (best system-design template)
E7	Multi-Objective Weather Routing via Hybrid PSO, 2022	J. Ocean Univ. China DOI 10.1007/s11802-022-4709-8	Non-dominated-sorting PSO	Route+fuel+risk models	Pareto-optimal route sets	Open ocean, not ice-specific	□□□
E8	Multi-objective ship weather routing via improved NSGA-III, 2024	ScienceDirect DOI via S2452414X24000141	NSGA-III + A-DDQN comparison	ECMWF weather data	Generates Pareto fronts for cost/time trade-off	Open ocean; needs ice-risk term added for polar use	□□□
E9	Estimation & Optimization of Ship Fuel Consumption — review, 2026	arXiv:2602.21959	Review	—	Surveys EMO, GA-PSO hybrids, evolutionary weather routing for fuel	Good Phase-1 reading for fuel-model choice	□□□

F. Decision Support Systems, GIS & Risk Assessment

No	Paper	Venue / DOI	Antarctic?	Method	Key Result	Limitation	R
F1	"Satellite-Based Sea Ice Navigation for Prydz Bay, East Antarctica" (SatSINS), 2017	Remote Sensing 9(6), 518. mdpi.com/2072-4292/9/6/518	Yes	Multi-source remote-sensing fusion → nautical-route recommendation	Operational decision-support demonstrated for RV Xuelong transiting to Zhongshan Station	Directly analogous to the SIH deliverable	Ci sc ge sc

No	Paper	Venue / DOI	Antarctic?	Method	Key Result	Limitation	R
F2	"Vessel Safety Navigation Under the Influence of Antarctic Sea Ice," 2025	<i>J. Mar. Sci. Eng.</i> 13(7), 1267. DOI 10.3390/jmse13071267	Yes	Satellite ice density/thickness → risk-assessment + icebreaker-escort optimization	96.3% ice-type classification accuracy; 15% risk-precision improvement vs. conventional methods	GNSS-R based; needs field validation at scale	□
F3	"Southern Ocean ice charts at the Argentine Naval Hydrographic Service...", 2022	<i>Front. Mar. Sci.</i> DOI 10.3389/fmars.2022.971894	Yes	Operational ice-chart production + iceberg-chart safety-of-navigation workflow	Real operational practice for Weddell/Bellingshausen/Antarctic Peninsula	Manual/analyst-driven, not ML	□
F4	POLARIS review — "Making sense of Arctic maritime traffic using POLARIS"	Book chapter (ResearchGate)	No	Risk Index Outcome (RIO) framework + network analysis for fastest safe route	Standard IMO-endorsed operational risk framework	Arctic-designed; Antarctic operators use analogous but non-standardized schemes	□ (risk based)
F5	Vessel-risk-aware DSS, 2025	<i>Maritime Policy & Management</i> DOI 10.1080/03088839.2025.2562543	No	GIS-MCDA + advanced Dijkstra	Multi-stakeholder route decision support	Fisheries-focused case study, transferable framework	□
F6	Arctic navigation risk assessment using ICESat-2 + GIS-AHP, 2026	ScienceDirect DOI via S156984322600066X	No	AHP + SMCE multi-criteria GIS risk mapping	Monthly/annual risk maps by vessel ice class	Arctic; bathymetry+SAR fusion approach transfers well	□
F7	Integrating Regional Ice Charts + Copernicus Sea Ice for Navigation Risk, Alaska	arXiv:2512.11083	No	POLARIS RIO computed from AIS + ice charts, 8 real vessels	Demonstrates POLARIS operationalization against real AIS tracks	Alaska case; methodology directly reusable	□

(Supporting RL/path-planning literature for the "recommended models" comparison in Section 7 — DRL+DWA (Ocean Eng. 2023), hierarchical DQN for MASS (Springer 2019), RA-SHRL COLREGs-constrained RL (Neurocomputing 2021), RL-A*+LSTM-DWA fusion (PMC 2025) — is listed in full in Section 7 rather than repeated here, since these are general maritime-autonomy papers, not ice-specific.)

4. Top 10 Essential Papers

- Braakmann-Folgmann et al. (2023), *The Cryosphere* — U-Net iceberg segmentation.** Why it matters: the only paper in this set that solves Component 3 (iceberg detection) with Antarctic Sentinel-1 data, a released dataset, and released code (Zenodo). Method: standard U-Net binary segmentation trained on 143 manually-outlined images of 5 named giant icebergs. What to adapt: the exact preprocessing (backscatter normalization) and post-processing (connected-component extraction) pipeline. Remaining gap the SIH project could close: it maps *known, already-named* giant icebergs one at a time — it does not do open-set detection of new/smaller icebergs, which your system likely needs.
- Budge & Long (2018), *IEEE JSTARS* — BYU/NIC Antarctic Iceberg Tracking Database.** Why it matters: this is the dataset paper behind the main long-running Antarctic iceberg position record. Method: scatterometer backscatter tracking + interpolation. Dataset: positions since 1978 (interrupted) and continuously since 1992. What to adapt: use this as your ground truth for training/validating any drift or trajectory model. Gap: coarse temporal sampling (irregular, sometimes weekly) — insufficient for a "real-time" DSS without fusing in Sentinel-1/AMSR2.
- Wang et al. (2023), *GRL* — SIPNet.** Why it matters: purpose-built for the exact forecast window (subseasonal, 1–8 weeks) most useful for a research-vessel voyage planning horizon, and it is Antarctic. Method: pure DL model on historical SIC only, benchmarked against ECMWF, NCEP, GFDL-SPEAR, and a linear Markov model. What to adapt: their evaluation protocol (comparing against three operational dynamical centers, not just persistence) is the rigor bar the SIH team should match. Gap: doesn't ingest wind/SST, leaving predictive skill on the table per the D1–D6 oceanographic literature.
- ASICNet (2026), *Advances in Atmospheric Sciences* — HybridLoss sea-ice-edge constraint.** Why it matters: solves the specific, literature-documented Antarctic problem of marginal-ice-zone bias by adding a physically-motivated loss term. Method: DL model with combined pixel + ice-edge-distance loss. What to adapt: the HybridLoss concept is a cheap, architecture-agnostic addition worth trying on whatever backbone the team picks. Gap: only validated on austral summer.
- Andersson et al. (2021), *Nature Communications* — IceNet.** Why it matters: the most cited, most reproducible template for U-Net-based sea-ice forecasting; even though Arctic, it defines best practice (probabilistic 3-class output, calibration checking) that ASICNet/ANTSIC-U-Net later imported. What to adapt: the probabilistic-output framing (instead of a single deterministic map) directly gives your DSS the uncertainty estimate the problem statement's "risk assessment" needs. Gap: Arctic-trained; would need full Antarctic retraining,

not just fine-tuning, given different ice-ocean-atmosphere regime.

6. Martin & Gallego (2025), arXiv/NeurIPS — 48-model optical-flow drift benchmark. Why it matters: the most rigorous existing benchmark for Component 2 (ice drift), with ground truth from GNSS buoys, not just self-consistency. Method: standard computer-vision optical flow networks applied to SAR. What to adapt: their evaluation metric (endpoint error against buoys) and their finding that off-the-shelf optical-flow networks transfer reasonably well to SAR — meaning a small student team does not need to design a bespoke drift architecture from scratch. Gap: Beaufort Sea (Arctic) only; needs Southern Ocean RADARSAT-2/Sentinel-1 validation, and no buoy network in Antarctica is as dense.

7. RouteView 2.0 (2025), Environmental Modelling & Software. Why it matters: this is the closest thing in the literature to a working end-to-end "prediction → route" decision-support system, i.e., a design template for the whole SIH platform, not just one module. Method: improved A* over a layered, time-varying risk graph built from SIC/SIT/temperature fields. What to adapt: its layered-graph architecture (separating the static topology from the time-varying ice-risk cost) is a very implementable pattern for the SIH capstone. Gap: Arctic Northeast Passage; an Antarctic version needs a POLARIS-style cost function since RIO, not raw SIC, is the internationally recognized risk metric south of 60°S too.

8. "Satellite-Based Sea Ice Navigation for Prydz Bay, East Antarctica" (SatSINS), 2017, Remote Sensing. Why it matters: the single most directly analogous prior system — an actual Antarctic research-vessel (RV *Xuelong*) route decision-support tool built from multi-source remote sensing, with no equivalent regulatory ice-service (unlike the Arctic). What to adapt: their catalog of which remote-sensing/met product is useful at which spatial/temporal scale is a ready-made data-sourcing checklist for NCPOR's own transits. Gap: pre-dates modern deep learning; case-study scale (single bay, single ship).

9. "Vessel Safety Navigation Under the Influence of Antarctic Sea Ice" (2025), JMSE. Why it matters: the most recent Antarctic-specific paper combining satellite ice classification, near-real-time risk scoring, and icebreaker-escort optimization in one framework — essentially a blueprint for Phases 5–8 of your roadmap. Method: satellite-derived ice density/thickness → risk model (GNSS-R-based classification, 96.3% accuracy) → escort optimization. What to adapt: their risk-precision improvement metric (15% over "conventional methods") gives you a concrete number to beat/match. Gap: relies on satellite latency; not a continuous real-time system yet.

10. Holland & Kwok (2012), Nature Geoscience — Wind-driven trends in Antarctic sea-ice drift. Why it matters: the landmark physical-oceanography paper establishing why Antarctic sea ice moves the way it does (wind forcing dominates), which every drift/trajectory model — ML or physical — implicitly needs to respect as a sanity check. Method: statistical analysis of a 19-year satellite ice-motion record. What to adapt: use their wind-forcing relationship as a physics-informed feature or as a baseline your ML drift model must beat. Gap: purely diagnostic/statistical, not predictive — it tells you *what matters*, not how to forecast it operationally.

5. Research Gap Analysis

(Each gap below is traceable to a specific paper above — none invented.)

- Antarctic data scarcity relative to Arctic.** The large majority of DL sea-ice-forecasting papers (A1, A6, A8, A9, A10, A11) are Arctic-trained; genuinely Antarctic DL forecasting papers exist (A2, A3, A4, A5) but are far fewer and mostly 2023–2026 — the sub-field is young. Gap: *limited benchmarked Antarctic architectures at daily/short-term resolution (most Antarctic DL work targets sub-seasonal/seasonal, per A2–A4).*
- Marginal-ice-zone / ice-edge bias.** Explicitly documented in A2 and A4 as a persistent, unsolved problem, even with the newest constrained-loss methods. Gap: *no paper in this set fully resolves ice-edge accuracy at daily resolution for Antarctica.*
- Lack of multimodal fusion at the iceberg-scale.** C1 and C2 use SAR-only inputs; none of the iceberg-detection papers found fuse ocean/wind covariates into the *detection* step (only into physical drift models, C4/C8). Gap: *no ML iceberg detector in this review ingests concurrent ocean-state data.*
- No unified real-time Antarctic iceberg-tracking + drift-prediction pipeline.** C6/C7 provide historical/periodic databases; C8's physics model is single-iceberg case-study; B1's drift-estimation benchmark is Arctic-only. Gap: *no evidence in the literature of an operational Southern-Ocean system chaining detection → identity-tracking → forecast for icebergs the way RouteView (E6) does for sea ice.*
- Uncertainty estimation.** Only A1 (IceNet) among the forecasting papers explicitly produces calibrated probabilistic output; the Antarctic-specific models (A2–A5) report point forecasts with error metrics but not predictive distributions. Gap: *Antarctic SIC forecasts with genuine uncertainty quantification are largely absent.*
- Sparse ground truth for Antarctic drift validation.** B1 validates against a dense Arctic GNSS-buoy network; no equivalent buoy density exists in the Southern Ocean in the papers reviewed — B3/B4/C8 rely on scatterometer tracks and single named icebergs. Gap: *model validation for Antarctic drift is inherently weaker than Arctic due to sparser in-situ data.*
- Fragmented route-optimization ↔ prediction integration.** E1–E9 optimize routes assuming ice fields as *given inputs*; F1/F2/F6 build risk/DSS layers; but no single paper in this review couples an ML *forecast* of future ice state directly into a *forward-looking* (not just current-conditions) route optimizer for the Southern Ocean specifically. RouteView (E6) is the closest analog and is Arctic. Gap: *prediction-into-optimization integration, at Antarctic scale, is essentially unaddressed in peer-reviewed literature.*
- No standardized Antarctic risk index equivalent to POLARIS uptake in ML pipelines.** POLARIS (F4) is Arctic/IMO-designed; F2's Antarctic risk model uses its own bespoke scoring, F3 is manual analyst charts. Gap: *no consensus, ML-compatible Antarctic risk-scoring standard exists yet — the SIH team would need to either adapt POLARIS or document a justified alternative.*
- Fuel-efficiency objective under-weighted for polar-specific routing.** Most multi-objective routing papers with strong fuel modeling (E1, E7, E8, E9) are open-ocean or generic-Arctic; ice-specific routing papers (E4, E5, F1–F3) emphasize safety/risk over fuel. Gap: *no paper jointly and rigorously optimizes fuel + Antarctic ice-risk with a validated fuel-consumption model.*
- Reproducibility.** Several of the most Antarctic-relevant papers (A2, A4, F1, F2, F3) do not have confirmed public code/data (see Section 9) — a real gap for a team hoping to build directly on them versus needing to re-implement from the paper text alone.

6. Paper-to-SIH Requirement Mapping

SIH Requirement	Research Area	Best Paper(s)	Dataset	Recommended Model	Your Implementation
Forecast Antarctic SIC (short-term, days)	A	A5 (ConvLSTM), A7 (methodology)	NOAA/NSIDC CDR (25 km) or higher-res AMSR2	ConvLSTM (fast to train, proven ~1-month Antarctic skill)	Daily SIC map + confidence band for the voyage corridor

SIH Requirement	Research Area	Best Paper(s)	Dataset	Recommended Model	Your Implementation
Forecast Antarctic SIC (sub-seasonal/seasonal)	A	A3 (SIPNet), A2 (ASICNet), A4 (ANTSIC-UNet)	NSIDC CDR + ERA5 covariates	U-Net w/ HybridLoss-style ice-edge term	1–6 month outlook for voyage planning window
Predict ice drift / motion	B	B1 (optical-flow benchmark), B2 (joint CNN)	Sentinel-1/RADARSAT SAR pairs, ERA5 wind	Pre-trained optical-flow net (e.g., RAFT-family) fine-tuned on SAR	Vector drift field overlay on the nav chart
Detect & track icebergs	C	C1 (U-Net), C3 (ViT+XGBoost)	Sentinel-1 SAR + BYU/NIC + circum-Antarctic dataset (C7)	U-Net segmentation → size/ID matching against C6/C7 records	Iceberg layer with ID, size class, last-seen position
Predict iceberg trajectory	C	C8 (physics + sea-ice force), B4 (analytical drift model)	ERA5 wind + Copernicus currents + live SIC	Hybrid: physical drift model as baseline + ML residual correction	24–120 hr iceberg position forecast with uncertainty cone
Ingest ocean/met covariates	D	D5 (ERA5), D6 (GLORYS12V1), D1 (SST ML)	ERA5, Copernicus Marine GLORYS12V1	Direct covariate ingestion; optional D4-style bias-correction network	Feature stack feeding modules above
Optimize route (risk-aware)	E, F	E1 (3D-ACA), E6 (RouteView 2.0), F4 (POLARIS)	Forecast SIC/SIT + POLARIS-style RIO cost	A* / Dijkstra over time-varying risk graph (RouteView-style) with POLARIS RIO as edge cost	Recommended track with alternates ranked by risk
Optimize route (fuel-aware)	E	E7/E8 (multi-objective PSO/NSGA-III), E9 (review)	Ship performance curve + weather	NSGA-II/III multi-objective search over {risk, time, fuel}	Pareto-front route options presented to operator
Decision-support / GIS front-end	F	F1 (SatSINS), F2 (Antarctic risk+escort)	All of the above, fused	Rule/threshold layer + map UI (GIS)	Operator dashboard: forecast layers + recommended route + risk score

7. Recommended ML/DL Approaches

Approach	Best used for	Input	Output	Pros	Cons / compute	Feasible for SIH timeframe?
CNN (plain)	Baseline SIC/iceberg classification	Gridded SIC or SAR patches	Class/label or coarse map	Simple, fast, well-understood	Limited long-range spatial context	☐ Yes — good starting baseline
U-Net	SIC forecasting (A1–A5); iceberg segmentation (C1)	Gridded multi-channel SIC/SAR	Full-resolution map/mask	Proven Antarctic + Arctic track record; moderate compute; strong literature support	Needs reasonably large labeled set for segmentation	☐ Yes — primary recommendation for both SIC forecasting and iceberg mapping
LSTM	Point/regional SIC or SST time series	1D time series per grid cell/region	Next-step value	Very cheap, easy to implement/debug	Ignores spatial structure unless combined with CNN	☐ Yes, as a quick regional baseline
ConvLSTM	Short-term (daily→1 month) spatiotemporal SIC forecasting (A5)	Sequence of SIC maps	Sequence of forecast maps	Captures both space+time; demonstrated Antarctic skill	Slower to train than plain CNN; error accumulates over long horizons	☐ Yes — second core recommendation
ConvLSTM + Attention / Transformer (SICFormer, MSS-STFormer, A10–A11)	Longer-horizon, multi-variable, super-resolution forecasting	Multi-channel spatiotemporal stack	High-res forecast + explainability maps	State of the art on Arctic benchmarks; explainability tools exist (SHAP/LIME per A10)	High compute/data hunger; A12 shows transformers can miss long-term seasonal cycle	☐ Optional stretch goal, not the core deliverable
Vision Transformer (ViT)	Feature extraction for iceberg classification (C3); high-res forecasting (A9)	SAR patches / SIC maps	Class or coarse forecast (paired with diffusion for super-res)	Strong performance on C3's Antarctic-adjacent task	Needs more data/compute than CNN/U-Net; C3 pairs it with XGBoost specifically to stay lightweight	☐ Feasible only in the ViT-features+XGBoost pattern (C3), not full ViT training from scratch
CNN-LSTM (hybrid)	SST/current time series with spatial context	Spatiotemporal ocean fields	Forecast map/series	Middle ground between ConvLSTM and plain LSTM	Architecture choices less standardized	☐ Yes, as ocean-covariate forecaster (D-series methodology)

Approach	Best used for	Input	Output	Pros	Cons / compute	Feasible for SIH timeframe?
Temporal Fusion Transformer	Multi-horizon forecasting with mixed static/dynamic covariates	Ship+route+weather features	Multi-step point+quantile forecast	Good uncertainty output; interpretable variable importance	Not seen used in the ice-forecasting papers reviewed here — would be a novel application	△ Interesting but unproven in this domain; medium risk
Graph Neural Network (GNN)	Route-graph-structured forecasting (A11) or maritime traffic graphs	Grid-as-graph or port/route network	Node-level SIC/SIT, or route-graph edge costs	Naturally represents the routing graph itself	Fewer polar-specific GNN papers found (mostly port-traffic GNNs, not ice-routing)	△ Use for the <i>routing graph</i> structure, not yet proven for ice-field forecasting specifically
Random Forest / XGBoost	Iceberg classification (C3); tabular risk scoring	Engineered features (backscatter stats, climate params)	Class label / risk score	Cheap, interpretable, small-data-friendly	Not spatial by itself — needs feature engineering	□ Yes — strong, low-risk recommendation for the risk-scoring/classification layer
Reinforcement Learning (DQN/PPO)	Local collision-avoidance / dynamic obstacle path planning	Ship state + local obstacle map	Steering/speed action	Handles dynamic, reactive avoidance well (E4-style, DRL+DWA)	Reward-shaping is hard; large training time; sim-to-real gap for polar RL	△ Feasible only as a <i>local</i> module on top of a global A*/graph route — not as the whole routing system
A* / Dijkstra (classical)	Global route search over a cost graph	Static/dynamic cost grid (ice risk, fuel)	Optimal/near-optimal path	Deterministic, explainable, fast, exactly what RouteView 2.0 (E6) and POLARIS-based DSS (F5) use operationally	Needs a good cost function (this is where your ML forecasts plug in)	□ Yes — primary recommendation for the global route-search layer
Multi-objective optimization (NSGA-II/III, PSO, ACO)	Balancing risk vs. time vs. fuel (E1, E5, E7, E8)	Candidate routes + cost components	Pareto-optimal route set	Directly matches the PS's "safe and fuel-efficient" requirement	More complex to tune/evaluate than single-objective A*	□ Yes — recommended as the layer <i>on top of</i> A* for final route ranking

Bottom line for a student SIH team: U-Net (SIC forecasting + iceberg segmentation) + ConvLSTM (short-term temporal dynamics) + XGBoost/Random Forest (iceberg classification & risk scoring) + A*/Dijkstra with a POLARIS-style, forecast-driven cost function + a lightweight multi-objective layer (NSGA-II) on top for fuel/time/risk trade-offs is the combination best supported by the literature at a reproducible, hackathon-feasible compute budget. Transformers, ViTs-from-scratch, and RL should be positioned as *stretch goals*, not the core deliverable, per A12's documented Transformer limitations and the compute burden noted throughout Section 7.

8. Trusted Dataset Recommendations

Dataset	Variable(s)	Source Org	Spatial Res.	Temporal Res.	Coverage	Period	
NOAA/NSIDC CDR Passive Microwave Sea Ice Concentration (G02202, v4/v5)	SIC	NOAA / NSIDC	25 km	Daily & monthly	Both poles	1978/1987–present	NetCDF
Near-Real-Time NSIDC CDR (G10016)	SIC	NOAA / NSIDC	25 km	Daily	Both poles	Recent (gap-filler to NetCDF present)	
Copernicus Marine GLORYS12V1 (GLOBAL_MULTIYEAR_PHY_001_030)	Currents, SST, SSS, SSH, sea-ice conc./thickness/velocity	Copernicus Marine Service (CMEMS)	1/12°	Daily/monthly	Global	1993–present	NetCDF
ERA5 Reanalysis	Wind, pressure, temperature, waves, etc.	ECMWF / Copernicus C3S	0.25° (~31 km)	Hourly	Global	1940–present	NetCDF
Sentinel-1 SAR (C-band)	Radar backscatter	ESA / Copernicus	~10–40 m (mode-dependent)	~6–12 day revisit (polar more frequent)	Global incl. Southern Ocean	2014–present	SAFE/C
SCAR International Iceberg Database	Iceberg positions, size class (ship-based obs.)	Norwegian Polar Institute / SCAR / AAD	Point observations	Per-cruise	Circum-Antarctic	1984/85–2010/11	Tabular
BYU/NIC Antarctic Iceberg Tracking Database	Iceberg positions (scatterometer)	Brigham Young Univ. / US National Ice Center	Point tracks	Weekly–irregular	Circum-Antarctic	1978, 1992–present	Text/CS

Dataset	Variable(s)	Source Org	Spatial Res.	Temporal Res.	Coverage	Period	
Circum-Antarctic Icebergs Dataset (2018–2023)	Iceberg polygons, size	ESSD / Sun Yat-sen Univ. (GEE pipeline)	From Sentinel-1	Annual (October snapshot)	South of 55°S	2018–2023	Vector/
SCAR Antarctic Digital Database (ADD)	Coastline, ice shelves, bathymetry, stations	British Antarctic Survey / SCAR	Vector, EPSG:3031	Updated ~6-monthly	Antarctica (to 60°S)	Ongoing	Shapefi
Polar View (BAS operational service)	Sea-ice charts, iceberg tracking, AMSR2 SIC	British Antarctic Survey	Varies by product	Near-real-time	Southern Ocean + Arctic	Ongoing	Web vie

All datasets above are publicly accessible (registration may be required for NSIDC/Copernicus). ERA5 and GLORYS12V1 require a free CDS/Copernicus Marine account; Sentinel-1 requires a free Copernicus Data Space account.

9. Reproducibility Analysis

Paper	Data public?	Code public?	Architecture described in enough detail?	Methodology reproducible?	Metrics clearly reported?
A1 IceNet	☐ (obs+CMIP6, standard)	☐ (Nature Comms supplementary + known GitHub follow-ups)	☐	☐	☐
A2 ASICNet	☒ (not verified — check paper for data-availability statement)	☒ (not verified)	☐	☒	☐
A3 SIPNet	☒ (not verified)	☒ (not verified)	☐	☒	☐
A4 ANTSIC-UNet	☒ (not verified)	☒ (not verified)	☐	☒	☐
B1 Optical-flow drift benchmark	☐ (RADARSAT-2, buoys — check access terms)	☒ (not verified — check arXiv listing)	☐	☐	☐
C1 Braakmann-Folgmann U-Net	☐ (Sentinel-1, public)	☐ Data+animations on Zenodo (10.5281/zenodo.7875599)	☐	☐	☐
C6 BYU/NIC database	☐	N/A (dataset paper)	☐	☐	☐
C7 Circum-Antarctic dataset	☐ (Sentinel-1/GEE, method described)	☒ (not verified)	☐	☐	☐
E6 RouteView 2.0	☒ (proprietary "Big Earth Data" pipeline — likely ☐ for raw system)	☐ (operational system, not open-sourced per available info)	☐ (algorithm described)	☒ (algorithm yes, full system no)	☐
F1 SatSINS	☒ (case-study specific inputs)	☐ (not an open-source tool)	☐	☒	☐
F2 Antarctic Vessel Safety / risk+escort	☒ (not verified)	☐ (not verified)	☐	☒	☐
D5 ERA5 / D6 GLORYS12V1	☐	N/A (data products)	N/A	☐	N/A

Legend: ☐ Available · ☒ Partially available / not verified — check source directly · ☐ Not available.

10. Recommended System Architecture

flowchart TD
 subgraph Data["Data Ingestion Layer"]
 S1[Sentinel-1 SAR]
 AMSR[AMSR2 / NSIDC CDR SIC]
 ERA5[ERA5 Wind/Pressure/Temp]
 GLORYS[Copernicus GLORYS12V1 Currents/SST]
 ICEDB[BYU-NIC + SCAR Iceberg Records]
 end
 subgraph Predict["Prediction Layer"]
 SIC_M[U-Net / ConvLSTM: SIC Forecast]
 DRIFT_M[Optical-Flow Net: Ice Drift Field]
 ICE_DET[U-Net Segmentation: Iceberg Detect/Track]
 ICE_TRAJ[Physics Model + ML Residual: Iceberg Trajectory]
 OCEAN_M[CNN-LSTM: SST/Current Nowcast]
 end
 subgraph Risk["Risk & Fusion Layer"]
 RIO[POLARIS-style Risk Index Outcome]
 XGB[XGBoost: Ice-Type / Hazard Classification]
 end
 subgraph Route["Route Optimization Layer"]
 GRAPH[Time-Varying Graph: RouteView-style]
 ASTAR[A* / Dijkstra: Candidate Routes]
 MOO[NSGA-II/III: Fuel-Time-Risk Pareto Front]
 end
 subgraph DSS["Decision Support / GIS Front-End"]
 MAP[GIS Map: SCAR ADD basemap + live layers]
 DASH[Operator Dashboard: recommended route + risk score + confidence]
 end
 S1 --> ICE_DET
 S1 --> DRIFT_M
 AMSR --> SIC_M
 ERA5 --> SIC_M
 ERA5 --> DRIFT_M
 GLORYS --> ICE_TRAJ
 GLORYS --> OCEAN_M
 ICE_DET --> ICE_TRAJ
 ICE_DET --> ICE_DET
 ICE_TRAJ --> ICE_DET
 ICE_TRAJ --> SIC_M
 RIO --> ICE_DET
 RIO --> ICE_TRAJ
 XGB --> ICE_TRAJ
 XGB --> RIO
 XGB --> RIO
 XGB --> GRAPH
 XGB --> ASTAR
 XGB --> MOO
 XGB --> DASH
 GRAPH --> ASTAR
 GRAPH --> MOO
 GRAPH --> DASH
 ASTAR --> MOO
 ASTAR --> DASH
 MOO --> DASH
 DASH --> ICE_DET
 DASH --> ICE_TRAJ
 DASH --> MAP
 DASH --> DASH

Why this shape: it mirrors the only two papers in the literature that actually deliver *end-to-end* systems (RouteView 2.0, E6, and SatSINS, F1) — prediction outputs are converted into a **graph cost**, not consumed directly by the route search, and a **POLARIS-style risk index** (F4) sits between raw predictions and the optimizer so the routing layer stays swappable/auditable.

11. Research Roadmap

Phase 1 — Literature: Read (in order) E3 (Arctic weather-routing review, for vocabulary/taxonomy) → C1 (Braakmann-Folgmann, for a concrete reproducible DL pipeline) → A1 (IceNet, for the U-Net/probabilistic-forecast template) → F1 (SatSINS, for the closest existing Antarctic DSS) → F4 (POLARIS, for the risk-scoring standard) → E6 (RouteView 2.0, for the system architecture template).

Phase 2 — Data: Register for NSIDC Earthdata, Copernicus Data Space (Sentinel-1), Copernicus Marine Service (GLORYS12V1), and Copernicus Climate Data Store (ERA5). Pull the BYU/NIC and SCAR iceberg records for historical validation. Scope the Antarctic sector(s) and season(s) matching NCPOR's actual voyage routes (e.g., Prydz Bay analog, per F1).

Phase 3 — Preprocessing: Reproject everything to a common polar-stereographic grid (EPSG:3031, matching SCAR ADD); co-register SAR imagery with SIC grids and ERA5/GLORYS12V1 at matching timestamps; build labeled iceberg-segmentation masks (following C1's manual-outline protocol) for any Antarctic Sentinel-1 scenes not already covered by C7's dataset.

Phase 4 — Model Development: Build the U-Net SIC forecaster (A1/A5 template) and the U-Net iceberg segmenter (C1 template) first, since both have the clearest reproducible precedent. Add ConvLSTM temporal extension once the static U-Net baseline works. Add the XGBoost/VIT-features iceberg classifier (C3 template) as a second, complementary detector.

Phase 5 — Prediction: SIC model → daily-to-monthly concentration maps with confidence bands; drift model → vector ice-motion field; iceberg pipeline → detected icebergs with ID, size class, and a forward trajectory cone (physics model per C8/B4, optionally ML-corrected).

Phase 6 — Route Optimization: Convert prediction outputs into a POLARIS-style RIO cost surface (F4/F7 template); build the time-varying graph (RouteView-style, E6); run A*/Dijkstra for candidate routes; run NSGA-II/III on top for the fuel-time-risk Pareto front (E1/E7/E8 templates).

Phase 7 — Validation: Benchmark SIC forecasts against persistence and, if feasible, a public seasonal product, using IIEE/MAE (as in A3/A7); benchmark drift against BYU/NIC or any available buoy/GPS-tagged iceberg track (B1/C8 protocol); benchmark iceberg detection with F1-score/IoU against manually labeled scenes (C1 protocol); benchmark routes qualitatively against historical NCPOR/known Antarctic vessel tracks where available (F3-style operational sanity check).

Phase 8 — Decision Support System: Integrate all outputs into a GIS front-end over the SCAR ADD basemap; expose forecast layers, iceberg tracks, and ranked route options with their risk/fuel/time trade-offs to the operator, following the dashboard pattern implied by F1/F2/E6.

12. Final "Must Read First" List (5–10 papers)

1. Braakmann-Folgmann et al. (2023) — *The Cryosphere* — U-Net iceberg mapping (has open code+data)
2. Andersson et al. (2021) — *Nature Communications* — IceNet (the reference DL sea-ice architecture)
3. Wang et al. (2023) — *GRL* — SIPNet (Antarctic subseasonal SIC, matches your forecast horizon)
4. "Satellite-Based Sea Ice Navigation for Prydz Bay" (2017) — closest existing Antarctic DSS precedent
5. RouteView 2.0 (2025) — *Environmental Modelling & Software* — end-to-end system architecture template
6. Budge & Long (2018) — IEEE JSTARS — BYU/NIC Antarctic Iceberg Tracking Database (your validation ground truth)
7. Holland & Kwok (2012) — *Nature Geoscience* — foundational Antarctic ice-drift physics
8. POLARIS review ("Making sense of Arctic maritime traffic using POLARIS") — the risk-scoring standard to adopt/adapt
9. "Vessel Safety Navigation Under the Influence of Antarctic Sea Ice" (2025) — *JMSE* — most recent Antarctic risk+escort integration
10. "Optimal ice routing of a ship with icebreaker assistance" (2019) — *Ocean Engineering* — multi-objective ice-routing formulation

Verification note

Every source above was retrieved via live search and cross-checked against at least one independent hit (journal page, arXiv abstract, or press coverage) where possible. Items flagged "**not verified**" should be pulled up directly by the team before citing in a report, since either the search results did not expose the specific detail (e.g., a full author list or exact code-availability statement) or the claim came from a secondary citation rather than the primary source.

